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Numerical simulations of capacitive coupling in graphene nanodevices using NGSolve

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Vienna, February 28, 2025

Declaration of Authorship

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Maximilian Klammer

Kurzfassung

Die moderne Si-MOSFET-Technologie stößt an ihre fundamentalen Grenzen hinsichtlich exponentiellen Leistungswachstums und weiterer Miniaturisierung. Aufgrund seiner herausragenden elektrischen Eigenschaften hat Graphen großes wissenschaftliches Interesse als potenzielles Material zur Steigerung der Rechenleistung moderner Informationstechnologien geweckt. Ein zentraler Aspekt der Optimierung graphenbasierter Bauelemente liegt im Verständnis der Quantenkapazitätseffekte, die aufgrund der zweidimensionalen Struktur von Graphen entstehen. Diese Arbeit befasst sich mit der numerischen Simulation von Graphen- und GNR-FETs, um die Mechanismen der kapazitiven Kopplung zu untersuchen und unsere Ergebnisse mit der bestehenden Literatur [1] zu vergleichen, um die Gültigkeit unserer Simulationsmethode zu bestätigen. Die Simulationen werden mit dem multiphysikalischen Finite-Elemente-Solver NGSolve in einer Python-Umgebung durchgeführt. Zunächst analysieren wir einen unendlichen Graphenkanal und validieren unsere numerische Simulation durch den Vergleich mit analytischen Modellen. Anschließend untersuchen wir endliche GNR-Kanäle unterschiedlicher Breiten und beobachten eine umgekehrte Proportionalität zwischen der Tiefe des Quantenkapazitätsdips und der GNR-Breite. Durch die Implementierung metallischer oder graphenbasierter Seitengates analysieren wir deren Abschirmungseigenschaften. Die komplexe Wechselwirkung zwischen Seiten- und Rückgate-Kopplung führt bei Graphen-Seitengates zu nichtlinearen Effekten, die eine effektiv negative Quantenkapazität zur Folge haben. Schließlich bestätigen wir die von Reiter et al. beobachteten Quantenkapazitätspeaks für Graphen-Seitengates.

Abstract

State-of-the-art Si-MOSFET technology is approaching its fundamental limits in terms of exponential performance growth and further miniaturization. With its exceptional electrical properties, graphene has garnered significant scientific interest as a potential material for enhancing the computational power of modern information technology devices. A crucial aspect of optimizing graphene-based device performance lies in understanding the quantum capacitance effects that emerge due to its 2D-structure. The topic of this thesis is the numerical simulation of graphene- and GNR-FETs to gain insights into the mechanisms governing capacitive coupling and compare our results with existing literature [1] to confirm the validity of our simulation method. The simulations are conducted using a multiphysics finite element solver, NGSolve, within a Python environment. First, we analyze an infinite graphene channel and establish the validity of our numerical simulation by comparison with analytical models. Next, for finite GNR channels of varying widths, we observe an inverse proportionality between the quantum capacitance dip depth and GNR width. By incorporating metallic or graphene sidegates, we examine varying screening properties for each material. The complex interplay between sidegate and backgate coupling for graphene sidegates introduces nonlinear effects which result in an effectively negative quantum capacitance. Finally, we are able to validate the quantum capacitance peaks reported by Reiter et al. for graphene sidegates.

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CHAPTER 1

Introduction

With the invention of the transistor in 1947 by J. Bardeen, W. Brattain and W. Shockley at Bell Labs the era of modern information technologies began. These small components, which control the current flowing between two contacts using a third one, are indispensable in modern information technologies due to their ability to amplify and switch electronic signals. Their applications range from telecommunication devices such as radios, televisions and mobile phones, where they amplify weak signals, to the computer industry, where they form the core components of microprocessors and memory chips.
A modern computer alone contains around ten millions [2] of these components. The mass production of the well-established silicon MOSFET (Metal-Oxide-Semiconductor Field-Effect Transistor) technology on silicon wafers combined with continuous miniaturization across eight orders of magnitude since the invention of the MOSFET [2] has led to incredibly low production costs and remarkable computational power. This continuous advancement in transistor scaling has been a key driving force behind the exponential growth in processing capabilities. The long-standing standard set by Moore's law, which predicted a doubling of the number of transistors on a semiconductor chip every two years, has recently been surpassed as further miniaturization becomes increasingly challenging and costly. Moreover, as we approach the atomic scale in device fabrication, with state-of-the-art technology now reaching 2 nm chips [3], fundamental physical limits will inevitably pose significant barriers to continued scaling.
Current research on enhancing transistor performance is exploring alternative materials, with graphene being a promising candidate. Due to its exceptional electrical and thermal conductivity graphene exhibits lower resistance and superior heat dissipation compared to silicon. These properties suggest a significant potential for improved efficiency and performance. Additionally, graphene's high carrier mobility makes it well-suited for high-frequency applications, while its 2D-structure enables ultra-sensitive sensing technologies. However, challenges such as integration with existing semiconductor fabrication processes and the absence of an intrinsic bandgap must be addressed before graphene can replace silicon in mainstream electronics.
While the absence of a bandgap in graphene can be addressed through the use of graphene nanoribbons (GNRs) or bilayer graphene, an important consideration in optimizing device fabrication is the influence of quantum capacitance effects in gate coupling mechanisms. These effects arise due to the low density of states (DOS) of graphene near the Dirac point. To achieve precise control over current flow in the channel, it is essential to understand how the device responds to variations in gate voltage.

CHAPTER 2

Theoretical Foundations

The following chapter gives an overview of the underlying theoretical background of this work. Firstly, graphene's material characteristics, first and foremost its electronic structure, is explained. Then in comparison the electronic characteristics of GNRs are introduced. As a prototypical example the working principle of MOSFET devices are established and on the basis of this GFETs and GNRs are introduced. Lastly, the important concept of quantum capacitance is presented briefly.

2.1 Graphene

Graphene is a monolayer of carbon atoms arranged in a hexagonal lattice. A stack of several of these layers forms the well-known 3D material graphite, from which the 2D material graphene is isolated. This was first achieved in 2004 by A. Geim and K. Novoselov via a relatively simple procedure using sticky tape, which is made possible due to the weak van-der-Waals forces between subsequent layers. Although this process is not industrially scalable it has facilitated research on graphene which has flourished in the recent past due to graphene's promising mechanical and electronic properties. The high charge mobility of up to $\sim 1 \times 10^6\,\mathrm{cm^2\,V^{-1}\,s^{-1}}$, combined with an optical transparency of $\sim 97.7\,\%$ and great mechanical strength and flexibility with a Young's modulus of $\sim 1\,\mathrm{TPa}$ [4] make graphene interesting for optoelectronic devices such as LCD displays. Furthermore, the high charge mobility, the property of near-ballistic transport at room temperature and a minimum conductivity of $\frac{4e}{h}$ near zero carrier concentration make it ideal for nanoelectronic devices [5] such as graphene field effect transistors (GFETs) or graphene quantum dots. Due to its big surface area and the thus increased reactivity graphene is also interesting for biological and chemical sensing applications.

2.1.1 Electronic structure

In its elemental form carbon's four valence atoms occupy the 2s and 2p orbitals. However, in a crystal one of the 2s electrons is excited to the $2p_z$ orbital due to interaction with neighboring atoms and the thus resulting reduction in total energy of the system. In graphene hybridization between the 2s and $2p_x$ and $2p_y$ orbitals forms three sp^2 orbitals [6]. The bonding between atoms in the planar graphene honeycomb structure is due to these orbitals along the plane, leading to strong covalent σ-bonds which are responsible for graphene's outstanding mechanical properties.

Orthogonal to this plane the $2p_z$ orbitals form relatively weak π-bonds with delocalized electrons, which are cause for the optical and electronic properties.

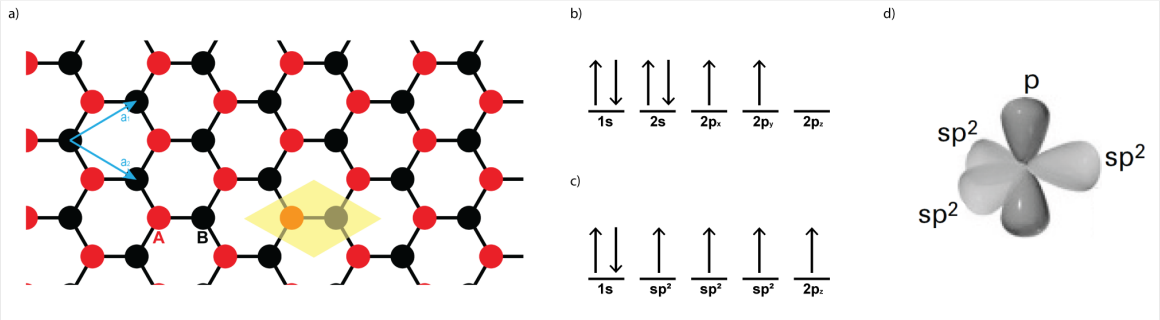


Figure 1: (a) Graphene lattice with basis vectors a_1 and a_2. The primitive unit cell is represented by the yellow rhombus. The hexagonal lattice consists of two sublattices of inequivalent carbon atoms A and B. (b) Electron configuration of elemental carbon and (c) graphene. (d) Graphene hybrid orbitals. Fig. d) taken from [6].

Graphene's electronic properties can be described via its band structure. An analytical expression can be derived by solving the time-independent Schrödinger equation using a nearest-neighbor tight binding (NNTB) ansatz and the electron-hole symmetry approximation for graphene which holds for energies close to E_F [6]. The solution for the π-bands of graphene and thus the conduction and valence band is

$$E(\vec{k})^{\pm} = \pm t\sqrt{1 + 4\cos\left(\frac{\sqrt{3}a}{2}k_x\right)\cos\left(\frac{a}{2}k_y\right) + 4\cos^2\left(\frac{a}{2}k_y\right)} \tag{2.1.1}$$

A commonly used value for the nearest-neighbor hopping potential t is 3.1 eV corresponding to a measured Fermi velocity of $v_F \approx 10^6\,\mathrm{m\,s^{-1}}$ [6]. The lattice constant of graphene a is the same for both lattice vectors and equal to $\sqrt{3}$ times the bonding length between neighboring carbon atoms, i.e. $a = |\vec{a_1}| = |\vec{a_2}| = \sqrt{3}a_{C-C} \approx 2.46\,\text{Å}$.

The unit cell of graphene consists of two carbon atoms (see fig. 1a) each of which has one electron in its p_z shell. The bonding (π) and antibonding (π^*) interactions between the p_z orbitals result in the formation of a valence band and a conduction band. There are N possible k-states in the valence band, where N is the number of primitive unit cells in the graphene layer. At zero temperature and considering the twofold spin degeneracy of each state the $2N$ electrons fill up the entire valence band and the conduction band is empty. Taking a look at the bandstrucure (see fig. 2) one sees that the highest occupied states lie in the high symmetry points K and K' of the Brillouin zone (BZ), hence per definition E_F lies there. Additionally, the two bands are in contact at these points (i.e. there is no bandgap) rendering graphene a semi-metal.

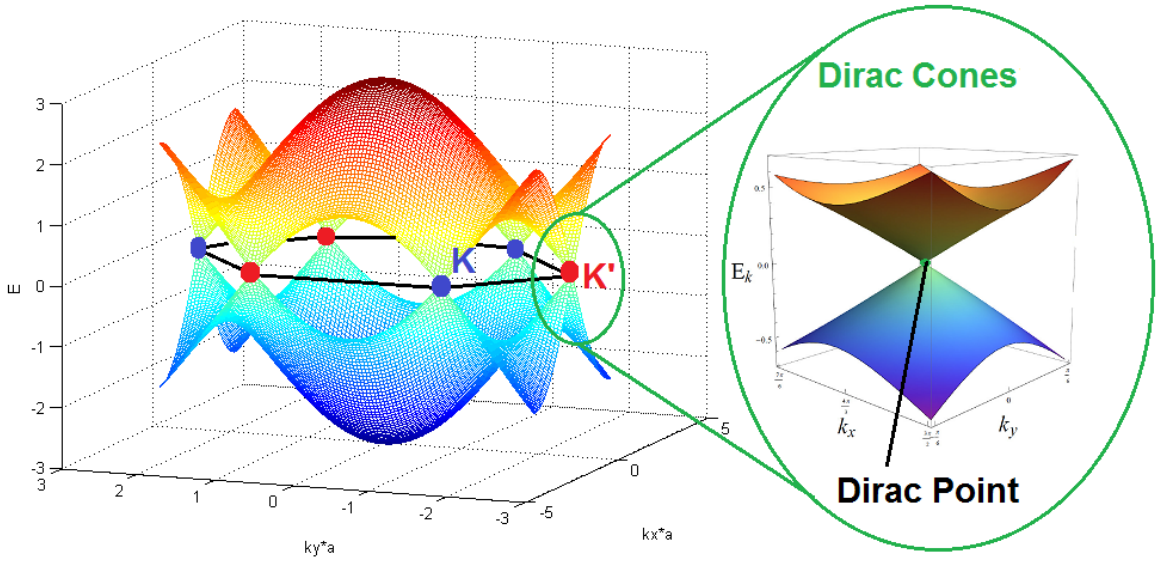


Figure 2: NNTB band structure of graphene. The hexagonal BZ gives rise to two distinct high-symmetry points K and K'. Zooming in at these points reveals a nearly linear behavior. Figure taken from [7].

In proximity of the high symmetry points K and K' of the BZ the band structure follows a simple linear relationship described by

$$E(\vec{k})^{\pm} = \pm v_F \hbar |\vec{k}| \tag{2.1.2}$$

Consequently, these points are called Dirac points because the effective mass of the charge carriers in this region is zero, and hence their behavior is described via the relativistic Dirac equation. Most of the electronic properties of graphene which are of interest for the use in nanoelectronic devices are determined by the behavior near these points and thus equation 2.1.2.

Another crucial concept for understanding a material's electronic properties is the density of states (DOS) which is defined as the number of states available for occupation by particles per unit volume of a system and per energy interval. The DOS can be calculated by dividing the differential area in k-space dA_k for the energy shell $[E(\vec{k}), E(\vec{k}) + \Delta E]$ of the band structure by the area of one k-state in the 2D k-space $\frac{(2\pi)^2}{\Omega}$, where Ω is the area of the system in real space.

$$g(E)dE = 2g_Z \frac{dA_k}{(2\pi)^2/\Omega} \tag{2.1.3}$$

The factor 2 comes from the spin degeneracy, another factor $g_Z = 2$ stems from the zone degeneracy which is due to the two independent K-points K and K' being related to each other by time reversal symmetry. Implementing these factors, using $dA_k = 2\pi k dk$ and normalizing the DOS to Ω gives

$$g(E) = \frac{2}{\pi}|k\frac{dk}{dE}| \tag{2.1.4}$$

The final expression for the DOS $g(E)$ is derived by using $k = \frac{E}{\hbar v_F}$ from Eq. 2.1.2.

$$g(E) = \frac{2}{\pi(\hbar v_F)^2}|E| \tag{2.1.5}$$

A linear relation for the DOS near the Dirac point is obtained for both negative energy values (holes as charge carriers) and positive energies (electrons). At the Fermi energy E_F the DOS becomes zero even though there is no bandgap. This is the reason for graphene being considered a semi-metal in comparison to metals that have no bandgap and a large DOS at the Fermi energy.

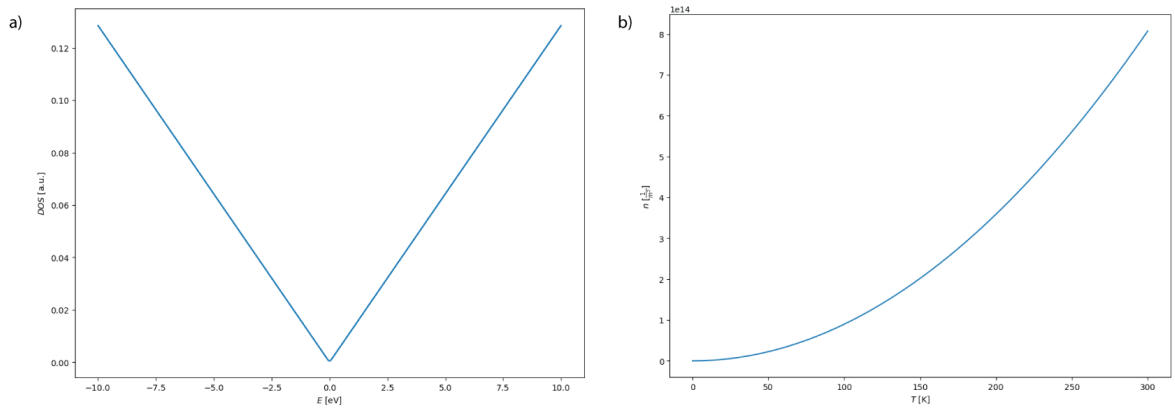


Figure 3: (a) Linear DOS of graphene near the Dirac point. The DOS becomes zero at the Fermi energy $E_F = 0$. (b) Charge carrier density of graphene. Note the parabolic dependence on temperature.

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The charge carrier density n is the number of charge carriers (electrons or holes) per unit volume in a material. For metals n is usually very high because the outer shell electrons contribute to the conduction band. In semiconductors n is typically much lower and depends on the band structure and temperature. In the following, the charge carrier density n of graphene will be derived. Mathematically, the charge carrier density is defined as

$$n = \int_{E_F}^{\infty} g(E)f(E)\,dE \tag{2.1.6}$$

where $g(E)$ is given by Eq. 2.1.5 and $f(E)$ is the Fermi-Dirac distribution defined as $f(E) = [1 + \exp(\frac{E-\mu(T)}{k_B T})]^{-1}$. For the chemical potential μ the approximation for room temperatures, i.e. $\mu(300\,\mathrm{K}) \approx E_F$, can be used. In thermal equilibrium, without external perturbation and using the analytical linear DOS approximation (Eq. 2.1.5) the integral yields

$$n = \frac{\pi}{6}(\frac{k_B T}{\hbar v_F})^2 \tag{2.1.7}$$

Unlike the exponential dependence of charge carrier density on temperature observed in semiconductors, graphene's intrinsic charge carrier density is proportional to T^2 due to its linear DOS and the absence of a bandgap.

2.2 Graphene nanoribbons (GNRs)

Because of the lack of a bandgap, graphene is not suitable for applications as a FET in logical computing. However, a bandgap can be opened by using either bilayer graphene and applying an interlayer potential to it or by using GNRs. Consequently, the focus will lie on the latter option.

GNRs are narrow stripes of graphene with widths up to tens of nanometers. Depending on the cross section at the edges, two different types of ideal GNRs can be differentiated: zigzag GNRs (zGNRs) and armchair GNRs (aGNRs). Both are semi-conducting with the bandgap depending inversely on the GNR width. The bandgap opens due to quantum confinement, i.e. the restriction of electron motion to one dimension, and due to arising edge states of the GNR. The generally more complex band structure can be calculated numerically using a Tight-Binding approach. The following expressions for energy dispersion, DOS and charge carrier density are applicable to all GNRs that exhibit a bandgap and were taken from [8].

Assuming isotropy of the electron and hole quasi-momenta in the graphene plane and aligning the x-axis along the infinitely extended axis of the GNR results in a quantized electron wave in y-direction because of the boundary conditions. Hence, the energy dispersion of the conduction band is split into n subbands.

$$E(n, k_x) = \hbar v_F \sqrt{k_x^2 + (\frac{n\pi}{w})^2} \tag{2.2.1}$$

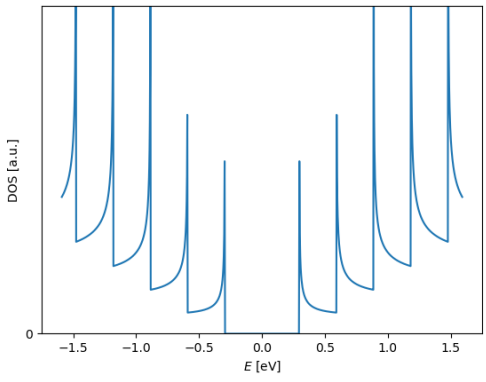


Figure 4: DOS of a 30 nm GNR near the Dirac point. A bandgap is opening around $E_F = 0$ which is inversely proportional to the width of the GNR. Van-Hove singularities can be observed at energies E_n where the group velocity $\frac{1}{\hbar}\frac{\partial E}{\partial \vec{k}}$ is zero.

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The total DOS of the conduction band is then the sum of the DOSs of all subbands

$$g(E) = \frac{4}{\pi\hbar v_F} \sum_n \frac{E}{\sqrt{E^2 - E_n^2}} \Theta(E - E_n) \tag{2.2.2}$$

where $E_n = n\pi\hbar v_F/w = nE_g/2$. Notice a bandgap of size E_g opening and van-Hove singularities at distinct energies E_n in the DOS (see Fig. 4). The 1D charge carrier density n in the case of $E_F/k_BT > E/k_BT \gg 1$ can be written as

$$n \approx \frac{4}{\pi\hbar v_F} \sum_n \sqrt{E_F^2 - E_n^2} \Theta(E_F - E_n) \tag{2.2.3}$$

When the Fermi energy E_F lies inside the bandgap the charge carrier concentration can be approximated by

$$n \approx n_i \exp(\frac{E_F}{k_BT}) \tag{2.2.4}$$

with $n_i \approx \frac{4}{w}\sqrt{\frac{\pi k_BT}{E_g}} \exp(-\frac{E_g}{k_BT})$ holding for gap sizes much bigger than the thermal energy, i.e. $E_g \gg k_BT$. This expression is similar to the one obtained for traditional semiconductors whereas Eq. 2.2.3 shows ripples due to the van-Hove singularities in the DOS arising from new transversal modes opening up, which are prominent when E_F lies outside of the bandgap.

In conclusion, GNR's promising electronic properties such as a high charge carrier mobility, a high current carrying capability and a tuneable bandgap makes them interesting for applications as small scale transistors in logical computing units.

2.3 Metal oxide semiconductor field effect transistor (MOSFET)

Transistors are electronic components usually made out of three electrical contacts. Their basic principle is to control the current flow between two contacts via the third one. This can be done by applying either a base current (bipolar junction transistor, BJT) or a gate voltage (field effect transistor, FET). Both may be used as amplifiers where a small change in V_{in} produces a large change in V_{out} or as electronic switches in digital circuits which can be in an on- or off-state. Today the MOSFET is by far the most common type of transistor and can be found in almost any modern electronics device. The structure and working principle of a n-channel MOSFET will be explained in the following.

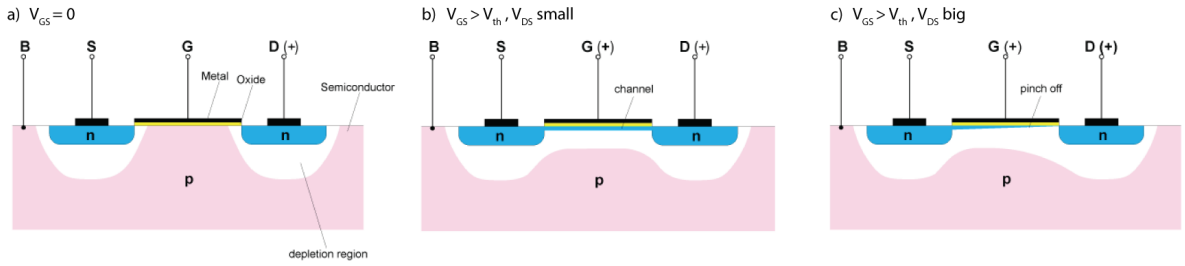


Figure 5: Structure and working principle of a n-channel MOSFET for (a) $V_{GS} = 0$, (b) $V_{GS} > V_{th}, V_{DS}$ small and (c) $V_{GS} > V_{th}, V_{DS}$ big. Figure adapted from [9]

For FETs current flow is controlled via an electric field which is created by applying a voltage to the gate electrode. The current flows in the channel between source and drain. Only one type of charge carriers, i.e. either electrons or holes, are responsible for the current in FETs in opposition to BJTs where the base current is comprised of both electrons and holes. The structure of a typical n-channel MOSFET is depicted in figure 5.

Source and drain are two separate n-doped areas, the substrate is p-doped. The gate electrode is deposited on an insulating oxide between the source and the drain. For no applied gate-source voltage $V_{GS} = 0\,\mathrm{V}$ the electrons would have to pass two oppositely polarized diodes to get through the channel from source to drain. Thus there is no current flowing in the channel (Fig. 5a). The application of a positive gate voltage results in a positive potential at the oxide-substrate interface that repels holes and attracts electrons in the substrate. Above the threshold voltage $V_{GS} > V_{th}$ a small channel of electrons is forming at the gate-substrate border. Applying a small positive voltage between drain and gate $V_{DS} > 0$ will allow electrons to flow from the source to the drain and thus the FET becomes conducting (Fig. 5b). For increasing V_{GS}, the channel gets bigger and hence the electron density within it is increasing too. However, if V_{DS} gets too positive the field of the gate electrode is overcompensated by the positive potential of the drain leading to a pinch-off of the channel and a saturation of the current (Fig. 5c) [9].

2.4 Graphene-FETs (GFETs) and GNR-FETs

Since the development of MOSFETs in 1960 by M. Atalla, D. Kahng and their team at Bell laboratories, the application of MOSFETs in integrated circuits (ICs) has followed Moore's law readily. G. Moore hypothesized in 1965 that the number of transistors fitting on an IC of fixed size will double roughly every two years. Lately, a physical limit has been reached for the further miniaturization of the proven Si-CMOS technology. Graphene's biggest advantage next to its high carrier mobility of $\approx 10^6\,\mathrm{m\,s^{-1}}$ might be the possibility to create monoatomic channels thus further decreasing MOSFET size.

The structure of a typical GFET is similar to a conventional MOSFET depicted in Fig. 5. However, there are some slight modifications possible. The top gate geometry shown in Fig. 5 can be used, the top gate can be omitted and replaced by a back gate below the Si substrate or a combination of top gate and back gate can be utilized. Its working principle is similar to that of a conventional FET, where an applied gate voltage controls the current flow in a channel. In a GFET this channel is a thin graphene layer deposited between the metal source and drain electrodes. Because of graphene's 2D nature the channel is only one atom thick resulting in current flowing only at the surface and making the device very sensitive. Unlike in Si-MOSFETs this current flow is ambipolar meaning that hole carrier conduction occurs in the channel under a negative bias and electron conduction for positive biases. For ideal graphene, i.e. no impurities or doping, the resulting conduction curves meet at the Dirac point.

The good electrical and thermal conductivity of graphene results in low resistance losses and better heat dissipation compared to Si. Using a special top-gate insulator material even facilitates the operation of the GFET at very high frequencies with extremely high switching speeds of up to 1 THz. In spite of that, the absence of a bandgap makes it difficult to turn off the transistor and thus unsuitable for applications in digital circuits. The ON/OFF-current-ratio of GFETs currently lies below 10 [10] which is too low for logic operations and magnitudes smaller than the ON/OFF-ratio of current Si-MOSFETs of 10^7.

While the application of wide-channel GFETs in rf-applications has been proven to surpass their Si-counterparts in certain aspects [10] they remain unsuitable for logic applications because of their poor ON/OFF-ratios due to the lack of a bandgap. For this reason research on GNR-FETs has been conducted with recent results for back-gated sub-10 nm-GNRs of ON/OFF-ratios of 10^6 compared to 10^7 obtained by current Si-MOSFETs [11]. One problem however remains the fabrication of 1D GNR channels as this usually requires chemical etching which proved itself to be very challenging.

The fact, that the DOS of graphene and GNR near the Dirac point is extremely low, i.e. the number of available states for electrons is much lower than in typical semiconductors, gives rise to interesting phenomena like quantum capacitance and an energy dependent and reduced screening. The weak screening near the Dirac point implies that small changes in the gate voltage V_{Gate} can cause significant shifts of the Fermi level E_F in the channel. This is a desirable feature for FETs because it enables precise control over the current flow.

But, because the efficiency of tuning E_F via V_{Gate} is not constant across the energy spectrum quantum capacitance related effects have to be considered when designing devices.

2.5 Quantum capacitance

Quantum capacitance (C_{QM}) is due to the finite energy needed to put more free charge carriers in the system because there is only a finite number of states available. Generally, quantum capacitance effects cannot be observed if the DOS is very large or the separation between adjacent energy levels is very small. In the devices examined in this study C_{QM} is in series with the geometric electrostatic capacitance. The expression for the total capacitance is given by equation 2.5.1, where $C_{cl} = \frac{\epsilon\epsilon_r A}{d}$ for a parallel plate capacitor and $C_{QM} = \frac{\partial Q}{\partial \phi}$. Q is the accumulated charge in the channel and ϕ the potential in the channel.

$$C_{tot} = \frac{1}{C_{cl}^{-1} + C_{QM}^{-1}} \tag{2.5.1}$$

In bulk materials C_{QM} is often very large because of the large number of electrons in the material that form a continuous band structure and thus are excitable by small energies. Hence, in bulk materials the electrostatic capacity dominates. In reduced dimensions however, the fewer number of electrons and the associated larger excitation energies result in a smaller C_{QM} which then is observable in C_{tot} [6].

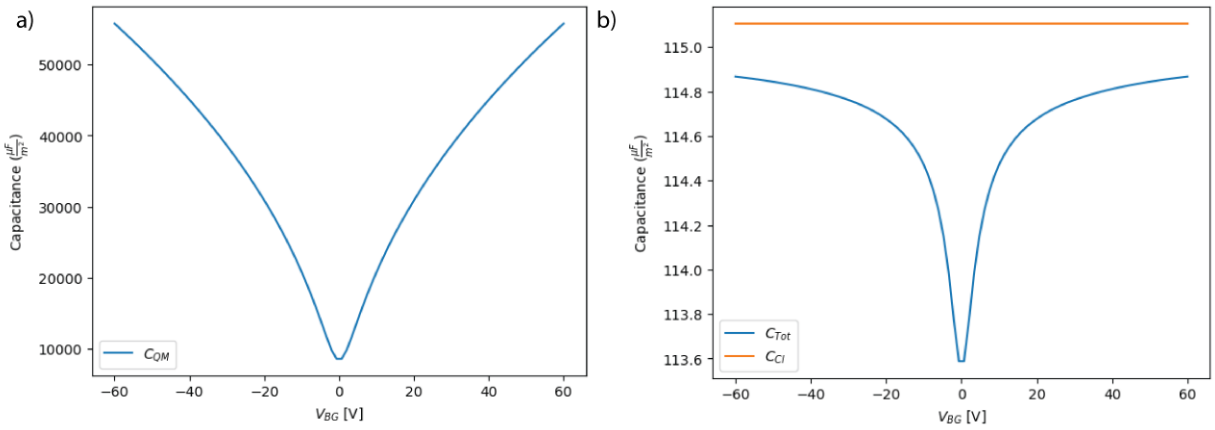


Figure 6: (a) C_{QM}, (b) C_{Cl} (orange line) and C_{Tot} (blue line) normalized to the channel area plotted over backgate voltage V_{BG} in $\mu F\,m^{-2}$.

C_{QM}, C_{Cl} and C_{Tot} of a device with infinitely extended graphene as channel can be seen in Fig. 6. The series connection of the classical capacitance C_{Cl} (orange line in Fig. 6b) and the quantum capacitance C_{QM} (blue line in 6a) lowers the total capacitance C_{tot} (blue line in Fig. 6b). At the Dirac point $V_{BG} = 0$, C_{QM} has a minimum because the DOS becomes minimal and a dip can be observed in C_{tot}.

CHAPTER 3

Simulation program

3.1 Analytical framework

The goal of our simulations is to obtain the free charge distribution ρ_f in the graphene/GNR channel of the examined devices. Having calculated ρ_f one is able to get the accumulated charge in the channel Q_{GNR}, which is needed for the calculation of quantum and total capacitance, via $C_{QM} = \frac{\partial Q_{GNR}}{\partial \phi}$ and $C_{tot} = \frac{\partial Q_{GNR}}{\partial V_{BG}}$ respectively.

To acquire ρ_f the non-linear Poisson equation (Eq. 3.1.1) has to be solved under Dirichlet boundary conditions. The non-linearity of the equation stems from the fact, that the inhomogeneity ρ_f also depends on the sought solution ϕ. Therefore, a self-consistency cycle has to be applied to obtain ϕ (see section 3.2.2).

$$\vec{\nabla} \cdot (\epsilon(\vec{r}, \phi)\vec{\nabla}\phi(\vec{r})) = -\frac{1}{\epsilon_0}\rho_f(\vec{r}, \phi) \tag{3.1.1}$$

The permittivity ϵ is modeled as a matrix with different permittivities in plane and out of plane due to the anisotropies occurring in the used materials. Generally, ϵ can also depend on the potential ϕ, e.g. in non-linear dielectrics. In the case of weak external fields it is possible to omit this dependence, which was done in our model. Additionally, spatial homogeneity was assumed therefore omitting the dependence on $\vec{r}$.

For a given potential $\phi(\vec{r})$ the local free charge density $\rho_f(\vec{r})$ can be calculated via a Thomas-Fermi approach according to Eq. 3.1.2.

$$\rho_f(\vec{r}) = e\int_0^{E_F(\vec{r})} DOS(E;\vec{r})\, dE \tag{3.1.2}$$

Using the linear approximation of the DOS of graphene yields the following local free charge carrier density at $T = 0$K.

$$\rho_f(\vec{r}) = \frac{eE_F^2(\vec{r})}{\pi(v_F\hbar)^2} \tag{3.1.3}$$

To consider finite temperature effects the Fermi-Dirac distribution has to be considered in the calculation of the local charge carrier density $\rho_f(\vec{r})$. Assuming $\mu(T) \approx E_F$ which is a good approximation at $T = 300\,\mathrm{K}$, Eq. 3.1.2 becomes

$$\rho_f(T;\vec{r}) = e\int_0^\infty DOS(E) f(T; E - E_F(\vec{r}))\, dE \tag{3.1.4}$$

where $f(T; E - E_F(\vec{r})) = [1 + \exp(\frac{E-E_F(\vec{r})}{k_B T})]^{-1}$ is the Fermi-Dirac distribution. Again using the linear approximation of the DOS of graphene one obtains the temperature-dependent local free charge carrier density [1].

$$\rho_f(T;\vec{r}) = \frac{2e}{\pi}(\frac{k_B T}{\hbar v_F})^2[Li_2(-e^{\eta}) - Li_2(-e^{-\eta})] \tag{3.1.5}$$

$Li_n(x)$ is the Polylogarithm and η is given by $\frac{E_F(\vec{r})}{k_B T}$. Eq. 3.1.5 is used for all simulations and is evaluated at $T = 300\mathrm{K}$.

To compare the results of our simulation with the literature, above all with [1], the actual GNR-DOS (Eq. 2.2.2) was approximated by the linear DOS of graphene (Eq. 2.1.5). A comparison of the two DOS is shown in Fig. 7a). Fig. 7b) represents the simulation results for C_{tot} obtained via the two different DOSs. They are in good agreement, except for ripples which are due to the van-Hove singularities in the GNR-DOS and a more-pronounced dip due to the bandgap in the GNR DOS.

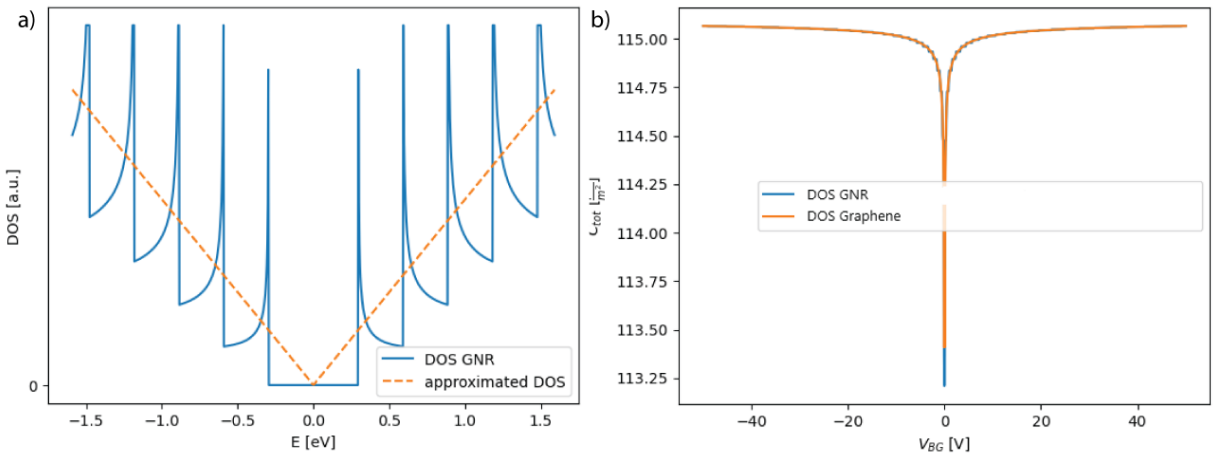


Figure 7: (a) Comparison of the DOS for graphene (orange dotted line) and a GNR (blue solid line). (b) Comparison of the simulated total capacitance C_{tot} using the DOS of graphene (blue line) and that of a GNR (orange line). The simulated GNR has a width of $w = 7\,\mathrm{nm}$.

3.2 Numerical framework

To translate the analytical expressions derived in the previous chapter into a numerical simulation some sort of discretization in the form of a mesh as well as a self-consistency cycle for solving the non-linear Poisson equation has to be implemented.

3.2.1 Discretization

To realize this discretization a high-performance, multiphysics finite-element software called NGSolve [12] was used. NGSolve can be used in problems ranging from solid mechanics to fluid dynamics to electromagnetics. With its integration with Python and a comprehensive set of pre-built finite element spaces, it enables flexibility, scalability, and efficient implementation. The procedure of solving the linear Poisson equation on a 3D geometry using NGSolve will be explained below.

$$-\vec{\nabla}^2\phi = f \tag{3.2.1}$$

The linear Poisson equation is represented by Eq. 3.2.1, where ϕ is the potential and the inhomogeneity f is given by ρ_f/ϵ_0. This equation is defined in a domain Ω with non-vanishing Dirichlet boundary conditions on a subset of the boundary $\Gamma_D \subset \partial\Omega$. The Dirichlet boundary conditions are given by

$$\phi\Big|_{\Gamma_D} = g \tag{3.2.2}$$

The boundary conditions g are the voltages applied to the different gates and Γ_D defines those areas of the gates where the boundary conditions are applied, e.g. $\Gamma_D = \Gamma_{BG} \cup \Gamma_{SG1} \cup \Gamma_{SG2}$. In the next step the weak form of the Poisson equation is derived through multiplication with a testfunction v_0 and integration by parts.

$$\int_\Omega \vec{\nabla}\phi \cdot \vec{\nabla}v_0 d\Omega = \int_\Omega f v_0 d\Omega \tag{3.2.3}$$

Splitting the potential ϕ into $\phi = \phi_0 + \phi_D$ where ϕ_D is an extension of the boundary conditions g into Ω computed by an NGSolve function the problem reduces from one with non-homogeneous boundary conditions to one with homogeneous boundary-conditions.

$$\int_\Omega \vec{\nabla}\phi_0 \cdot \vec{\nabla}v_0 d\Omega = \int_\Omega f v_0 d\Omega - \int_\Omega \vec{\nabla}\phi_D \cdot \vec{\nabla}v_0 d\Omega \tag{3.2.4}$$

Discretizing the domain Ω under the use of a suitable mesh and corresponding basis functions leads to a linear algebra problem for ϕ_0

$$A\phi_0 = f - A\phi_D \tag{3.2.5}$$

which is solved for ϕ_0 using a sparse solver algorithm. $A(\phi_0, v_0)$ is a bilinear form represented by the integral $\int_\Omega \vec{\nabla}\phi_0 \cdot \vec{\nabla}v_0 d\Omega$. The full potential ϕ is ultimately obtained by addition $\phi = \phi_0 + \phi_D$.

For more detailed information on the procedure refer to the NGSolve documentation [12]. Furthermore, a geometric and possibly more intuitive interpretation of this problem is given in [13].

3.2.2 Self-consistent solution

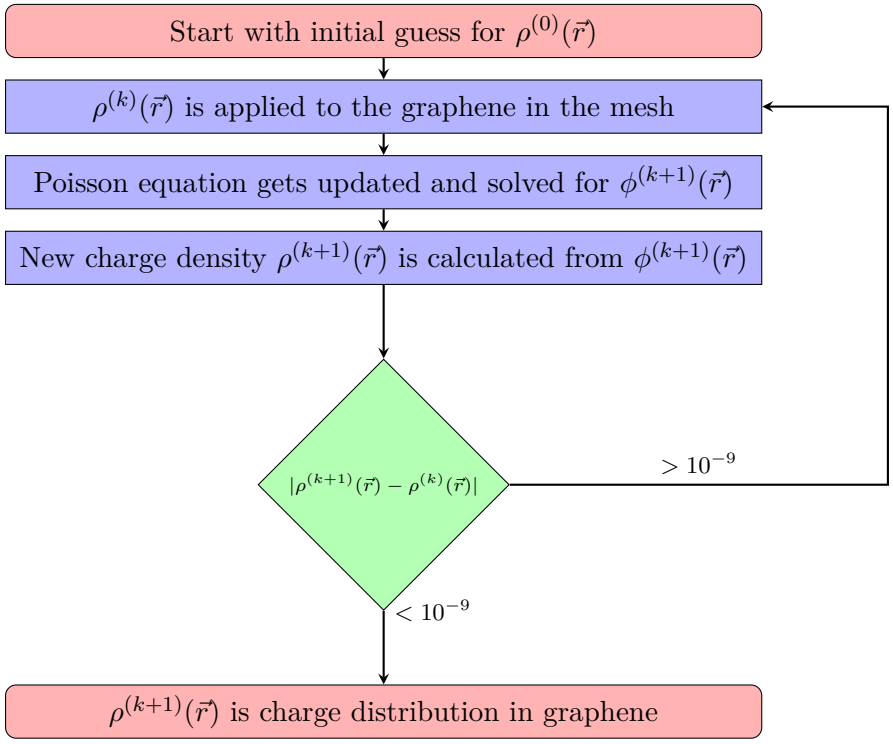


Figure 8: Schematics of the self-consistency cycle.

The self-consistency cycle starts with an educated guess of the local charge density $\rho^{(0)}(\vec{r})$ in the graphene channel, that is applied to the mesh. The non-linear Poisson equation (Eq. 3.1.1) then gets updated with this local charge density $\rho^{(0)}(\vec{r})$. The Poisson equation can now be solved for the potential $\phi^{(k+1)}(\vec{r})$ via the procedure described in the previous chapter. The newly calculated potential $\phi^{(k+1)}(\vec{r})$ sets the local fermi energy $E_F(\vec{r}) = e\phi(\vec{r})$ in the local DOS approximation and at zero temperature. Here, even though $T = 300\,\mathrm{K}$ we use the approximation $\mu(\vec{r}, T = 300\,\mathrm{K}) \approx E_F(\vec{r}) = e\phi(\vec{r})$, which is inserted into Eq. 3.1.5 to obtain the new local charge density $\rho^{(k+1)}(\vec{r})$. This cycle is repeated until the absolute difference between old and new charge density falls below a threshold of 10^{-9}. Then, the self-consistency cycle has converged and the local charge distribution $\rho^{(k+1)}(\vec{r})$ in the graphene channel has been found.

Thus, solving the non-linear Poisson equation amounts to finding the root of

$$\rho_n - \rho[\phi_{(n+1)}] = \rho_n - e \int_0^{e\phi_{n+1}} DOS(E, \vec{r}) dE \tag{3.2.6}$$

where the non-linear Poisson equation is given by $\vec{\nabla} \cdot (\epsilon \vec{\nabla} \phi_{(n+1)}(\vec{r})) = -\frac{1}{\epsilon_0} \rho_n(\vec{r}, \phi_n)$.

The accumulated charge in the graphene nanoribbon (GNR) Q_{GNR} can now be determined by integrating the local free charge carrier density $\rho_f(\vec{r})$ over the entire area covered by the GNR.

$$Q_{GNR} = \int_{GNR} \rho_f(\vec{r}) dV \tag{3.2.7}$$

Finally, the total capacitance C_{tot} of the device can than be calculated through differentiation by the backgate voltage $C_{tot} = \frac{\partial Q_{GNR}}{\partial V_{BG}}$.

3.3 Gate modeling

3.3.1 Metal gates

Due to the effectively infinite density of free charge carriers in metals, any applied voltage results in a uniform potential throughout the metal. Therefore, the potential in the gates is equal to the externally applied gate voltage and remains constant across the gate. This justifies the use of Dirichlet boundary conditions ($\phi = V_{Gate}$) in modeling the system. Dirichlet boundary conditions are used both in the modeling of the metallic back gate and in the modeling of metallic side gates.

3.3.2 Graphene gates

Graphene side gates (SGs) differ significantly from metallic side gates due to the unique electronic properties of graphene. Unlike metals, which have an effectively infinite density of states (DOS), the DOS in graphene is energy-dependent and vanishes at the Dirac point (see Fig. 3). This results in a highly tunable carrier density that must be taken into account when modeling the potential distribution in graphene SGs.

The potential and charge carrier distribution in graphene SGs are governed by the non-linear Poisson equation (Eq. 3.1.1). The charge density in graphene depends on the local DOS and the position of the Fermi energy E_F, which itself is influenced by the applied gate voltage (Eq. 3.2.2). To accurately model graphene SGs, a self-consistent approach to obtain the charge carrier concentration in the gate itself is necessary.

The process involves assuming an initial potential distribution given by the applied gate voltages, computing the carrier density using the energy-dependent DOS of graphene, solving the Poisson equation for the updated potential distribution, and iterating this process until convergence is achieved.

To correctly implement the applied gate voltages into the modeling process mixed boundary conditions are required. Dirichlet boundary conditions are applied at the contacts where the gate voltage is defined ($\phi = V_{Gate}$) and Neumann boundary conditions ($\frac{\partial \phi}{\partial n} = 0$) are applied at the vacuum-device interfaces, implying that ϕ does not change going beyond the boundary. This is also true for the modeling of metal SGs. Furthermore, the carrier concentration across the graphene side gate is determined based on the local potential, which is composed of the externally applied gate voltage and the self-consistently calculated electrostatic potential at each point. This iterative approach ensures an accurate representation of the coupled relationship between the potential and carrier density.

CHAPTER 4

Numerical Results and Discussion

In the following, the simulation results will be presented, interpreted and compared to the literature. The goal of this thesis was the recalculation of the geometrical structures simulated by Reiter et al. [1] and thus testing the suitability of the Poisson solver for problems of similar complexity. The examined capacitors consisted of a graphene or GNR channel, a metallic backgate (BG), SiO_2 as dielectric inbetween and optionally two sidegates (SGs) either metallic or made out of graphene. The results will be presented in order of increasing complexity of the geometry. Starting with idealized, infinite graphene the work will move on to simulate a device with a GNR channel of finite size. In the end, SGs will be introduced, first metallic SGs and lastly SGs made out of graphene. With increasing complexity new device-specific quantum-capacitance related effects will arise.

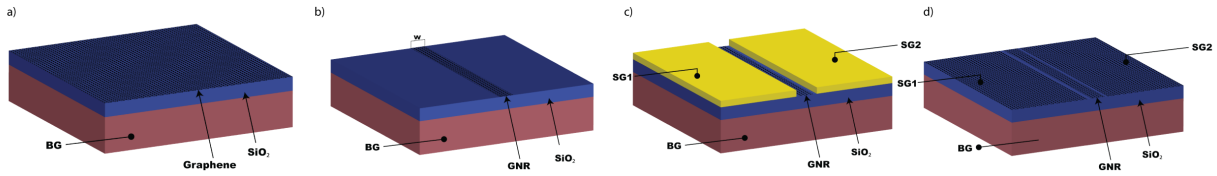


Figure 9: Geometry with (a) infinite graphene channel, (b) GNR channel of finite size w, (c) GNR channel of finite size w with metallic SGs each at distance t from the GNR and (d) GNR channel of finite size w with graphene SGs each at distance t from the GNR.

4.1 Infinite graphene

The examined device is depicted in Fig. 10. It consists of a metallic BG where a voltage is applied to alter the local charge density distribution in the infinitely extended graphene channel via capacitive coupling. The potential in the graphene channel is weakened by the SiO_2 dielectric between BG and channel in comparison to the potential at the BG. The uppermost layer of the BG forms a parallel plate capacitor with the graphene layer, the capacitance per unit area of which is given by $C_{Cl} = \frac{\epsilon_0 \epsilon_r}{d}$. The relative permittivity ϵ_r of the dielectric SiO_2 is taken from literature to be 3.9 and the dielectric thickness d is taken from [1] to be 300 nm.

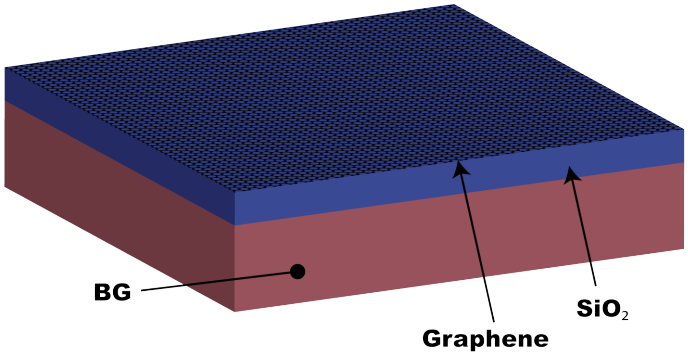


Figure 10: Capacitor formed by infinitely extended graphene sheet, SiO_2 dielectric and metallic BG.

The total capacitance C_{tot} of the device can be calculated via the following equation

$$C_{tot} = \frac{1}{C_{Cl}^{-1} + C_{QM}^{-1}} \tag{4.1.1}$$

where C_{Cl} is the previously mentioned, analytical expression for the classical parallel plate capacitor and C_{QM} can be calculated numerically by partial derivation of the accumulated charge in the graphene sheet $Q_{Graphene}$ with respect to the potential in the graphene sheet ϕ, i.e. $C_{QM} = \frac{\partial Q_{Graphene}}{\partial \phi}$.

The simulation results are presented in Fig. 11a). The total capacitance C_{tot} is normalized to the device area and plotted over varying BG voltage V_{BG}. The green dotted line represents the numerically simulated total capacitance and is compared to an analytical model (Eq. 4.1.2) shown as a solid blue line. Both results are in good agreement. Furthermore, the simulation results are in good agreement with the results obtained by Reiter et al., see Fig. 11b). For reference, the purely classical capacitance of the parallel plate capacitor formed by the graphene layer and the BG is also shown as an orange line.

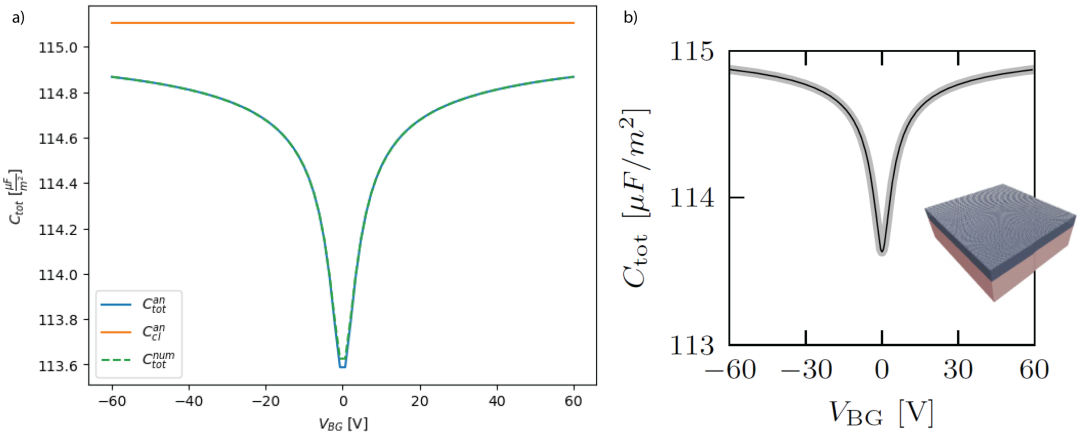


Figure 11: Comparison of (a) the simulation results from this work with (b) the simulation results by Reiter et al. [1] for a device with infinitely extended graphene channel.

The characteristic quantum capacitance dip (QC-dip) is clearly observed in Fig. 11. This phenomenon arises when the local potential within the graphene channel aligns with the Dirac point ($E_F = e\phi = 0$), where the density of states (DOS) vanishes. Owing to the symmetric design of the device, the Dirac point is uniformly achieved at a BG voltage of 0 V resulting in a narrow peak right there.

An analytical expression of the total capacitance C_{tot} can be found by using the following analytical expression [8] for the quantum mechanical capacitance C_{QM} in combination with the previously presented analytical expression for the classical capacitance C_{Cl} in combination with Eq. 4.1.1.

$$C_{QM} = \frac{2e^2 k_B T}{\pi(\hbar v_F)^2} \ln\left[2(1+\cosh\left(\frac{e\phi}{k_B T}\right)\right] \tag{4.1.2}$$

4.2 GNR without SGs

The device depicted in Fig. 12 features a graphene nanoribbon (GNR) with finite width w, positioned on top of a SiO_2 dielectric layer, which itself rests on a metallic BG.

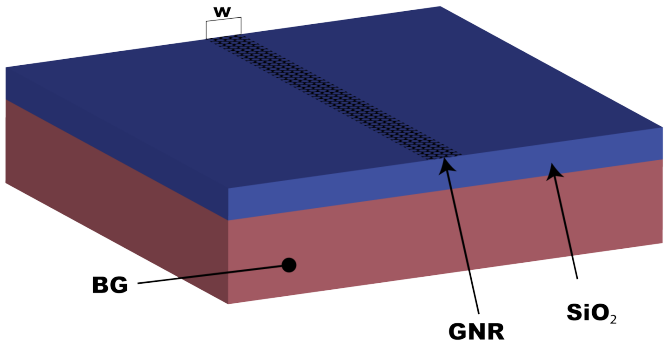


Figure 12: Capacitor formed by GNR of width w, SiO_2 dielectric and metallic BG.

An analytical expression for the classical capacitance C_{Cl} of this geometry can be found by assuming a homogeneous dielectric medium below and above the GNR and constant charge density on the GNR [14].

$$C_{Cl}(w) = C_0\left[\frac{2}{\pi}\arctan\left(\frac{w}{4d}\right) + \frac{w}{4d\pi}\ln\left(1+\frac{16d^2}{w^2}\right)\right]^{-1} \tag{4.2.1}$$

This width-dependent expression utilizes the classical capacitance of a parallel plate capacitor $C_0 = \frac{\epsilon_0 \epsilon_r}{d}$, modifying it through a functional expression. The total capacitance C_{tot} can again be calculated via Eq. 4.1.1, where C_{QM} is the previously mentioned equation for numerical calculation of the quantum mechanical capacitance.

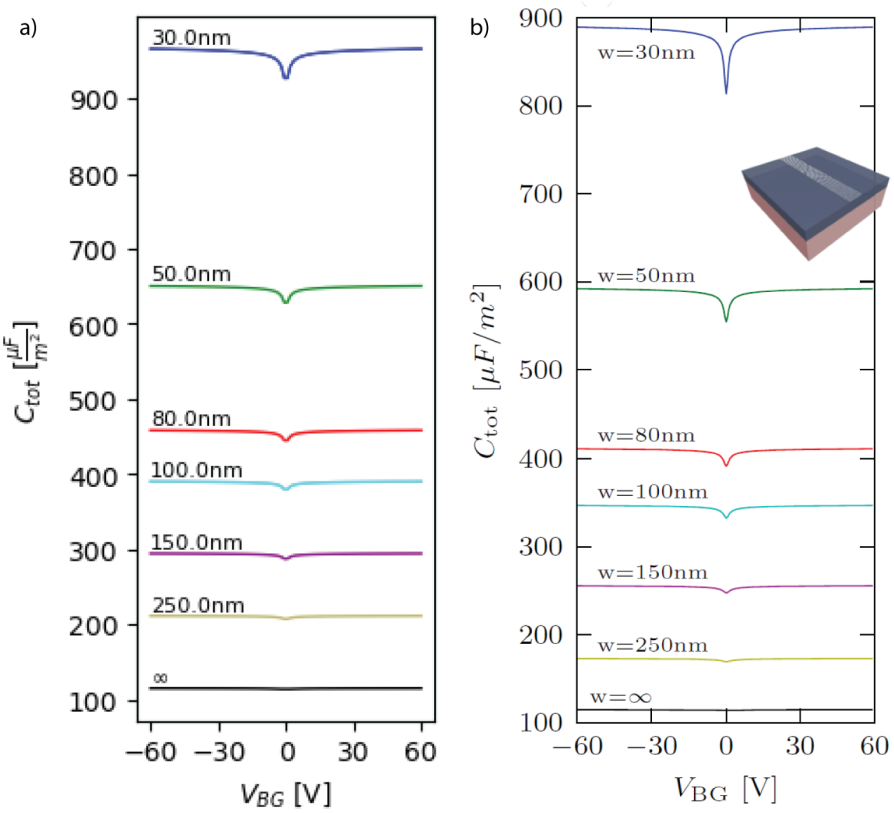


Figure 13: Comparison of (a) the simulation results from this work with (b) the simulation results by Reiter et al. [1], highlighting the effect of varying the GNR channel widths w.

The simulation results depicted in Fig. 13a) show the total capacitance per unit area C_{tot} plotted over the BG voltage V_{BG} for various GNR widths w. It can be seen, that a reduction in the GNR width results in a more pronounced QC dip. This effect can be attributed to the smaller DOS at reduced w. Additionally, as the width decreases, the density of field lines per area of GNR increases, leading to an increase in classical capacitance per unit area C_{Cl} hence increasing the total capacitance per unit area C_{tot} as well.

Fig. 13a) is in good qualitative agreement with the results from Reiter et al. shown in Fig. 13b). However, there are quantitative discrepancies due to the flawed assumptions made in the derivation of the analytical expression for the classical capacitance C_{Cl} (Eq. 4.2.1) used in this simulation. The expression for C_{Cl} assumes a uniform charge density throughout the GNR. But, this assumption is not fully satisfied in this scenario, leading to deviations from the theoretical predictions of Reiter et al. Furthermore, the derivation of C_{Cl} is based on the assumption of a homogeneous dielectric medium above and below the GNR, utilizing an effective dielectric constant $\epsilon_r = \frac{\epsilon_{SiO_2}+\epsilon_{Air}}{2}$. This approximation introduces additional discrepancies due to the in reality inhomogeneous dielectric environment and because this simulation uses $\epsilon_r = \epsilon_{SiO_2}$.

4.3 GNR with metal SGs

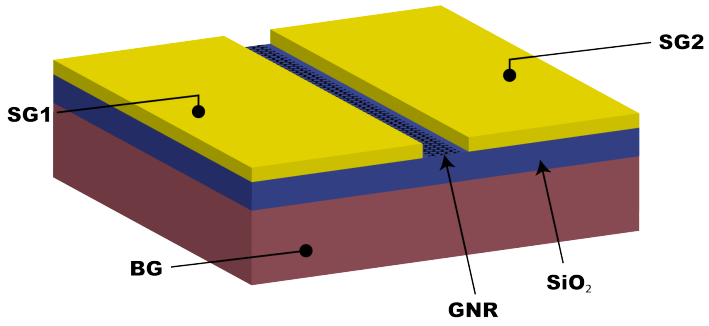


Figure 14: Capacitor formed by GNR channel of width w, SiO_2 dielectric, metallic BG and two metallic SGs with distance $t = 30\,\text{nm}$ to the GNR.

The examined device is shown in Fig. 14. To generate less computational effort all simulations were executed on a 2D cross section perpendicular to the GNR direction. This is plausible because the geometry is symmetrical along the GNR axis. For geometries including SGs no analytical expression exists for the classical capacitance C_{Cl}. Therefore, the total capacitance C_{tot} must be calculated directly using Eq. 4.3.1.

$$C_{tot} = \frac{\partial Q_{GNR}}{\partial V_{BG}} \tag{4.3.1}$$

This approach, however, introduces a small error which is illustrated in Fig. 15. For the infinite graphene geometry, for which we have established agreement between the analytical model and our numerical results (blue line) calculated via Eq. 4.1.1, the numerical results have been recalculated using 4.3.1 (blue line). A small yet not insignificant difference between the two lines can be seen. Calculating C_{tot} directly also means, that C_{QM} cannot be extracted from C_{tot} anymore as the inverse function theorem is no longer applicable to Eq. 4.1.1 without knowing C_{Cl}.

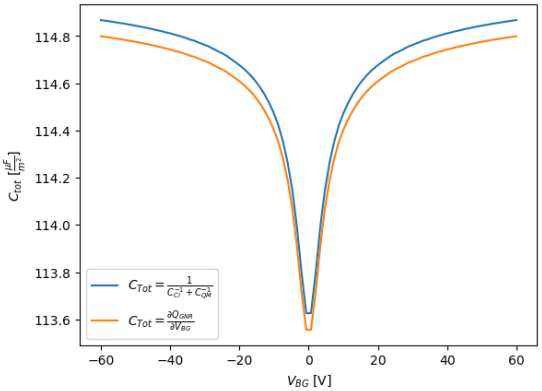


Figure 15: C_{tot} of infinite graphene geometry calculated via different methods.

4.3.1 Grounded metal SGs

The simulation results for grounded SGs, i.e. both SGs at zero voltage, can be seen in Fig. 16a). The classical capacitance C_{Cl} is not accessible in this simulation, thus the total capacitance per unit area C_{tot} is not normalized to C_{Cl} as in Fig. 16b) depicting the results from Reiter et al. In Fig. 16a), the total capacitance per unit area is plotted as a function of BG voltage for various GNR widths. A comparison with the corresponding results for the case without SGs, shown in Fig. 13a), reveals that both scenarios exhibit the same qualitative behavior, namely a more pronounced QC dip and an increase in C_{tot} for decreasing w.

The agreement with the results of Reiter et al. is not that obvious. As the width of the GNR decreases, the density of field lines per unit area increases, leading to a rise in C_{Cl} and, consequently, in C_{tot}. However, since Reiter's results are normalized by this width-dependent classical capacitance, the ordering of the results is reversed compared to ours. This reversal occurs because the rate of increase in C_{Cl} with decreasing width is greater than that of C_{tot}, where C_{QM} also plays a significant role.

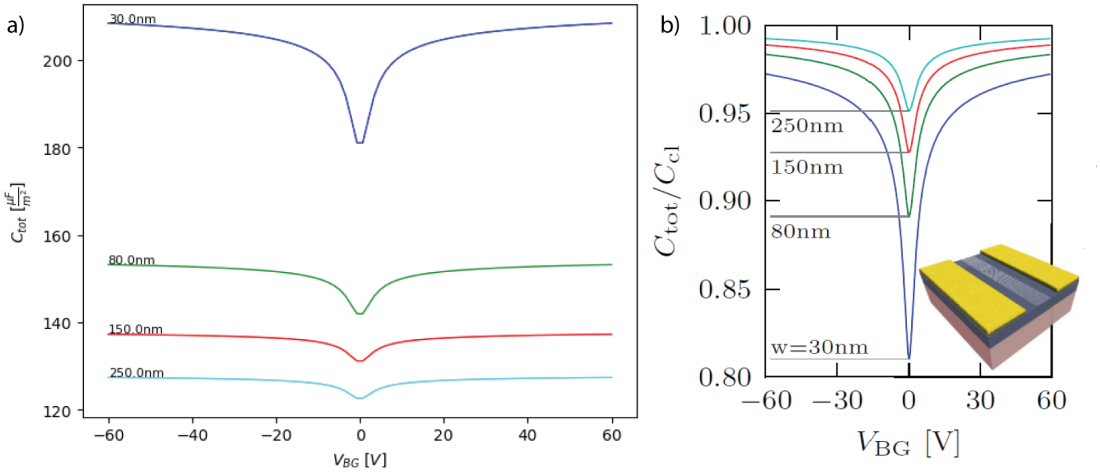


Figure 16: Comparison of (a) the simulation results from this work with (b) the simulation results by Reiter et al. [1], for grounded metal SGs and varying GNR widths w.

The total capacitance C_{tot} for the SG geometry is decreased compared to the geometry without SGs. This is because the introduction of SGs leads to screening effects. Without SGs the BG directly couples to the GNR through the dielectric SiO_2. When SGs are implemented, the field lines further away from the GNR no longer bend towards the GNR but towards the SGs. This means charge accumulates at the SGs and C_{tot} decreases. Additionally, the GNR itself has alternative pathways for field lines to terminate on the SGs rather than coupling directly to the BG, which further diminishes C_{tot}. This aspect is illustrated in Fig. 17.

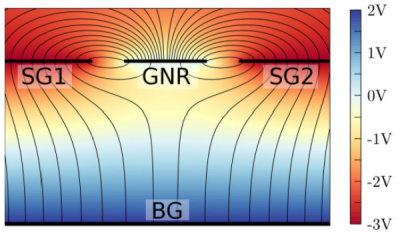


Figure 17: Electrostatic potential and field lines on 2D cross section perpendicular to GNR direction. Figure taken from [1].

4.3.2 Symmetric SG voltages ($V_{SG1} = V_{SG2}$)

In Fig. 18a) the numerically simulated total capacitance per unit area is plotted against the BG voltage for different symmetric SG voltages ($V_{SG1} = V_{SG2}$) ranging from 0 V to 7 V. The qualitative behavior of these results is in good agreement with the results from Reiter et al. shown in Fig. 18b). Again a quantitative comparison is not possible due to the fact that C_{Cl} is not obtainable in our simulation.

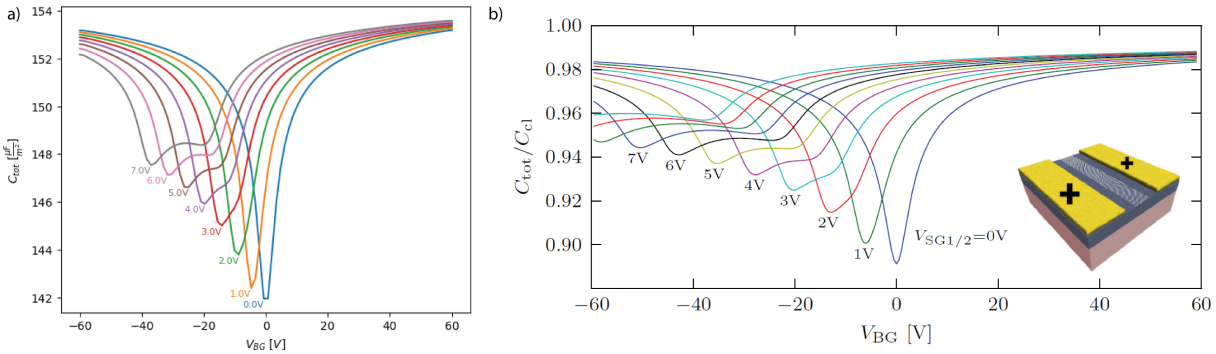


Figure 18: Comparison of (a) the simulation results from this work with (b) the simulation results by Reiter et al. [1], for the symmetric voltage configuration of metal SGs at $w = 80$ nm.

Two distinct phenomena occur in the simulation results. First, there is a progressive shift of the QC dip toward higher values with increasing applied V_{SG}. The QC dip occurs at the Dirac point, where $E_F = e\phi = 0$. For nonzero SG voltage, the condition $\phi = 0$ no longer coincides with a BG voltage of zero. In the case of positive V_{SG}, the position of the QC dip shifts to negative V_{BG}, where the influence of the SGs is compensated.

Second, the QC dip exhibits an asymmetric broadening, which becomes more pronounced at higher V_{SG}. This broadening is accompanied by an increasingly shallow QC dip. For nonzero SG voltages, a nonuniform potential distribution $\phi(\vec{r})$ develops across the GNR. As a result, different parts of the device reach their local Dirac point at different voltage configurations, causing the QC dip to become broader and less pronounced. An increase in V_{SG} leads to greater non-uniformity in the potential.

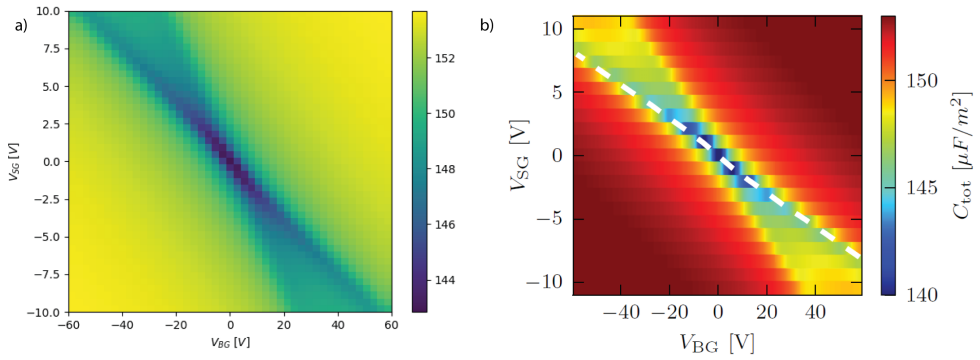


Figure 19: Contour plots of C_{tot} plotted over V_{BG} and V_{SG} for (a) the simulation results from this work and (b) the simulation results by Reiter et al. [1], for the symmetric voltage configuration of metal SGs at $w = 80\,\text{nm}$.

The contour plots in Fig. 19 highlight the behavior of C_{tot} with varying V_{BG} and V_{SG}. A quantitative comparison can be made between this work's results and the results from Reiter et al., where good agreement in the absolute value of C_{tot} is found. Moreover, their qualitative behaviors exhibit great resemblance. The shift of the QC dip can be seen by following the line of lowest values in C_{tot} in Fig. 19a) (darkest blue). Furthermore, the asymmetry and broadening of the QC-dip is observable in the widening of the blue area as the absolute value of V_{SG} increases.

An experimentally important quantity is the relative lever arm denoted by α_{rel}. It is defined as the ratio between the BG and the SG capacity (see Eq. 4.3.2), with $C_{BG} = \frac{\partial Q_{GNR}}{\partial V_{BG}}\Big|_{V_{SG}}$ and $C_{SG} = \frac{\partial Q_{GNR}}{\partial V_{SG}}\Big|_{V_{BG}}$ respectively. As the BG and SGs usually feature very different capacitive couplings due to the different spatial distances to the GNR, this quantity is important to consider when designing a GNR nanodevice.

$$\alpha_{rel} = \frac{C_{BG}}{C_{SG}} \tag{4.3.2}$$

Fig. 20 compares the results for α_{rel} of this simulation with the results of Reiter et al. This work's simulation (Fig. 20a) shows a similar qualitative energy dependence of α_{rel} compared to Reiter's simulation (Fig. 20b). An increase of α_{rel} of roughly 1 % relative to the average value can be found along a diagonal line spanning from the top left to the bottom right corner in Fig. 20a). Moreover, a decrease of again roughly 1 % is observed as a dark blue line crossing the yellow line at an angle. As Reiter states, that the experimentally observed α_{rel} is nearly a factor three larger than his calculation suggests [1], it is noteable that, although small, the difference between C_{tot} between our calculation and Reiter's has the correct trend. The residual discrepancy between simulation and experiment is most likely due to a significantly reduced SG coupling in the experiment because of charge accumulating on the rough edges of the GNR.

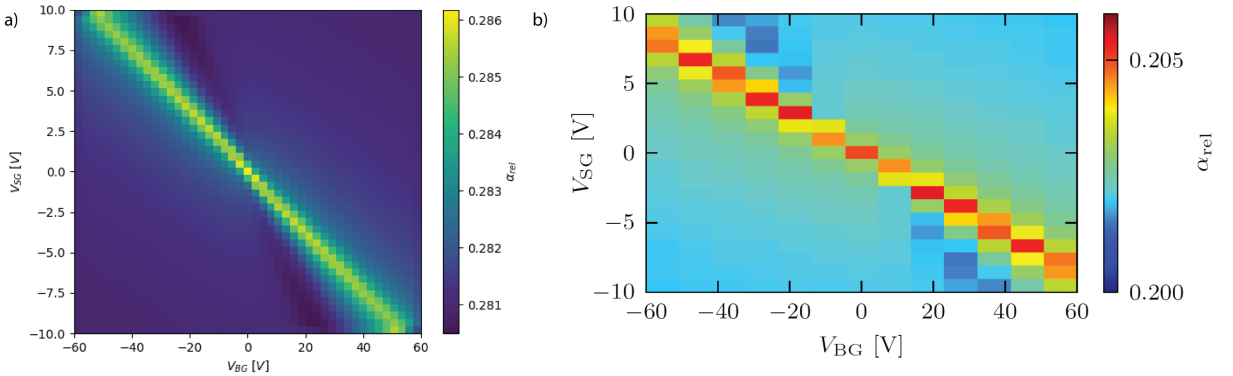


Figure 20: Contour plots of α_{rel} plotted over V_{BG} and V_{SG} for (a) the simulation results from this work and (b) the simulation results by Reiter et al. [1], for the symmetric voltage configuration of metal SGs at $w = 80\,\text{nm}$.

4.3.3 Antisymmetric SG voltages ($\mathbf{V_{SG1} = -V_{SG2}}$)

In Fig. 21a) the numerically simulated total capacitance per unit area is plotted against the BG voltage for different antisymmetric SG voltages ($V_{SG1} = -V_{SG2}$) ranging from 0 V to 7 V. The qualitative behavior of these results is in good agreement with the results of Reiter et al. shown in Fig. 21b). A quantitative comparison is not possible due to the fact that C_{Cl} is not available in our simulation.

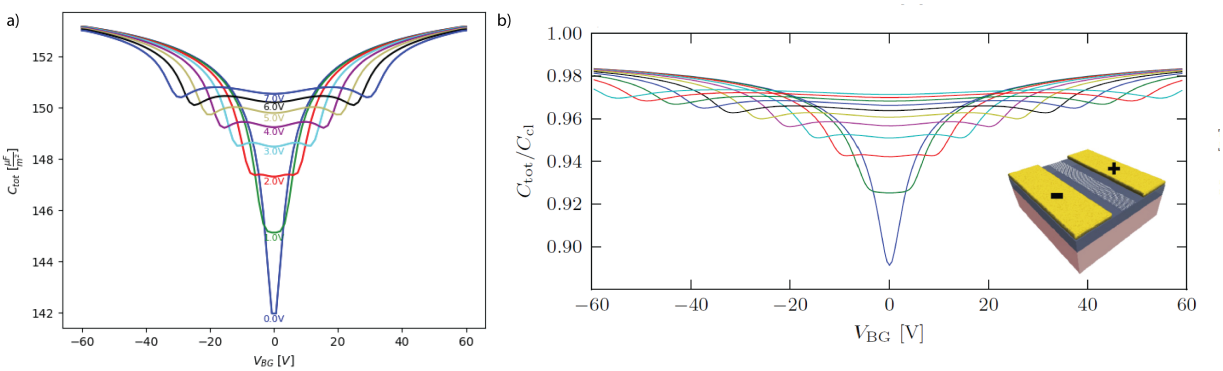


Figure 21: Comparison of (a) the simulation results from this work with (b) the simulation results by Reiter et al. [1], for the antisymmetric voltage configuration of metal SGs at $w = 80\,\text{nm}$.

For increasing V_{SG} a widening of the QC-dip occurs. This can be attributed to the lateral electric field across the GNR which arises because of the antisymmetry of the potential on the SGs. A non-uniform potential distibution $\phi(\vec{r})$ and thus a spatial variation of the Fermi level E_F throughout the GNR follows from this. Therefore, causing parts of the GNR to reach their local Dirac point at different BG voltages V_{BG}.

The contour plots in Fig. 22 again show great resemblance, although some features are more pronounced in Reiter's simulation in Fig. 22b). Furthermore, there is a small difference in C_{tot} when comparing the two results. The more pronounced QC-dip at low V_{SG} can be observed as the darkest blue parts of the contour plot in Fig. 22a) and the broadening of the QC-dip when increasing the absolute value of V_{SG} is represented by the blue cones.

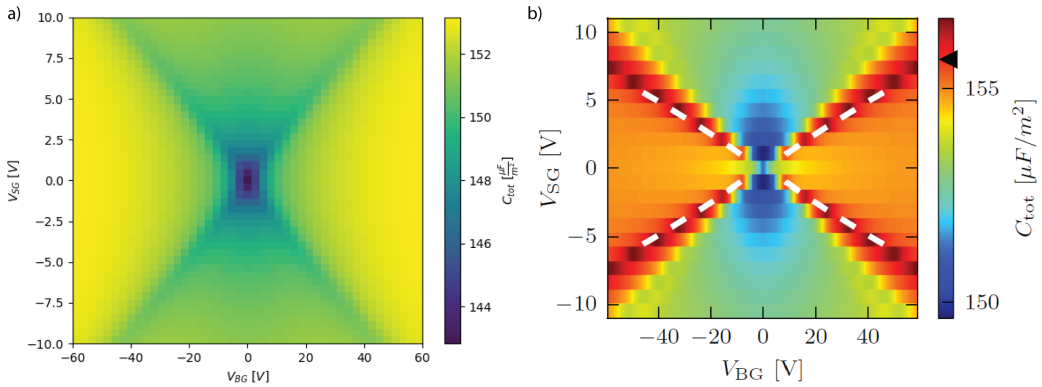


Figure 22: Contour plots of C_{tot} plotted over V_{BG} and V_{SG} for (a) the simulation results from this work and (b) the simulation results by Reiter et al. [1], for the antisymmetric voltage of metal SGs configuration at $w = 80\,\text{nm}$.

4.4 GNR with graphene SGs

The device depicted in Fig. 23 features a graphene nanoribbon (GNR) with finite width w, positioned on top of a SiO_2 dielectric layer, which itself rests on a metallic BG. The device is completed by two graphene SGs at a distance of 30 nm from the GNR. Again, this is simulated as a 2D geometry.

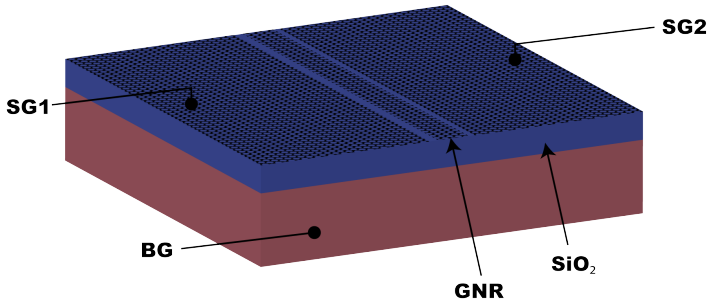


Figure 23: Capacitor formed by GNR channel of width w, SiO_2 dielectric, metallic BG and two graphene SGs with distance $t = 30\,\text{nm}$ to the GNR.

4.4.1 Grounded graphene SGs

This work's numerically calculated capacitance per unit area is plotted over the BG voltage for various GNR widths and both SGs at zero voltage. The results can be seen in Fig. 24a). Fig. 24b) illustrates the same results simulated by Reiter et al. and is normalized to the respective classical capacitance. Again, comparing this to Fig. 13a) for the geometry without SGs and to Fig. 16 for the geometry with metal SGs one can observe the same qualitative behavior. That is, a QC-dip, which is in its depth inversely proportional to the GNR width and an increase in C_{tot} as w decreases. The same is true for the results of Reiter et al. when considering the normalization to C_{Cl}. Since Reiter's results are normalized by this width-dependent classical capacitance, the ordering of the results is reversed compared to ours. This reversal occurs because the rate of increase in C_{Cl} with decreasing width is greater than that of C_{tot}, where C_{QM} also plays a significant role.

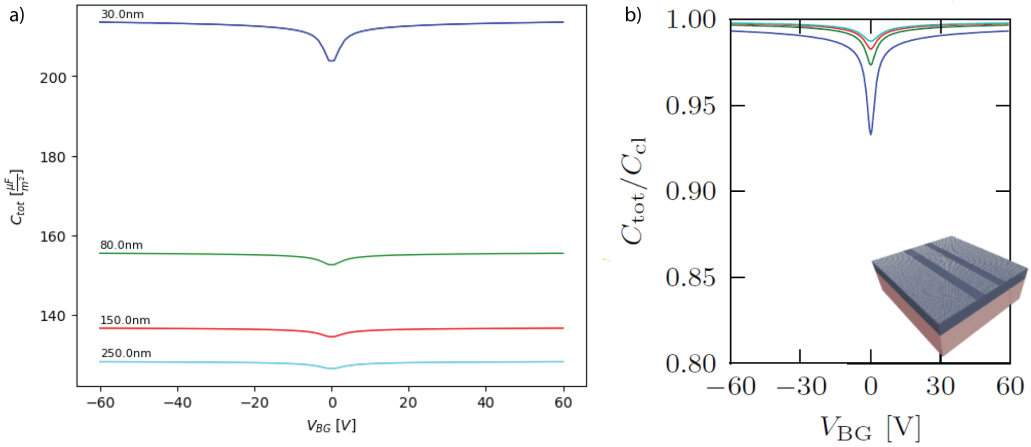


Figure 24: Comparison of (a) the simulation results from this work with (b) the simulation results by Reiter et al. [1], for grounded graphene SGs and varying GNR widths w.

A detailed analysis of the total capacitance for the three mentioned geometries reveals the relationship $C_{tot}^{NoSG} \gg C_{tot}^{Graphene} > C_{tot}^{Metal}$, driven by the different screening capabilities of the materials. Unlike metallic SGs, graphene SGs cannot accomodate an infinite amount of charge carriers, leading to a reduced screening effect. Consequently, the potential ϕ in the GNR is higher than in the geometry with metallic SGs but significantly lower than in the configuration without SGs, where no screening occurs. This reduced screening effect results in reduced BG coupling for the SG geometries, with the reduction being particularly pronounced in the case of metallic SGs.

Comparing these results with those obtained for metal SGs reveals that the QC dip is more pronounced in the latter configuration. This difference can be attributed to the emergence of effectively negative quantum capacitance in the case of graphene SGs, which gives rise to an additional QC peak. A more detailed discussion of this phenomenon is provided in Sec. 4.4.2 and the subsequent sections. For grounded SGs, the GNR and the SGs reach the Dirac point simultaneously. As a result, the QC peak overlaps with the larger QC dip, diminishing its depth.

4.4.2 Symmetric SG voltages ($V_{SG1} = V_{SG2}$)

Fig. 25a) shows the numerically calculated total capacitance per unit area plotted over the BG voltage for a device with a GNR width of 80 nm. The simulation results of Reiter et al. are observable in Fig. 25b), however, here the total capacitance is normalized to the classical capacitance, which was not possible in our simulation. Comparing our results qualitatively with the results of Reiter et al. shows a different behavior for positive SG voltages ($V_{SG1} = V_{SG2} > 0$). In our simulation, the dip shifts for positive SG voltages to negative BG voltages, while in Reiter's simulation exactly the opposite behavior occurs.

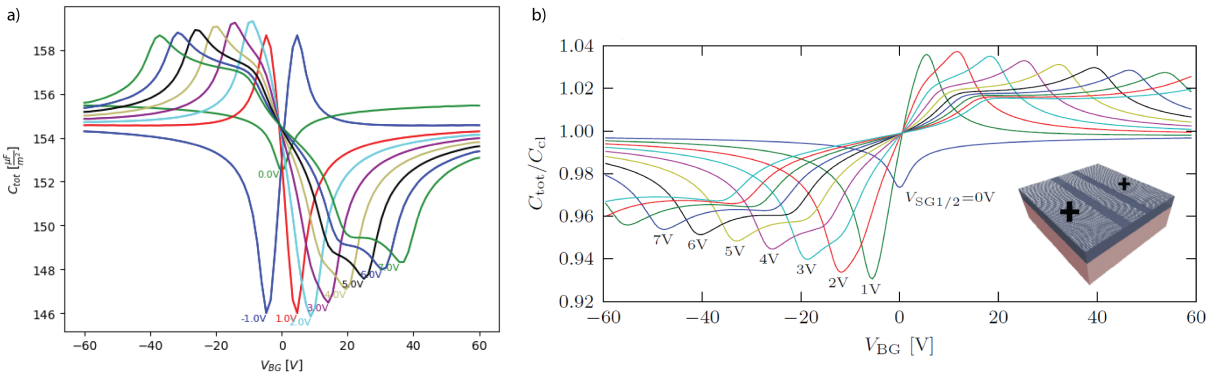


Figure 25: Comparison of (a) the simulation results from this work with (b) the simulation results by Reiter et al. [1], for the symmetric voltage configuration of graphene SGs at $w = 80$ nm.

When applying symmetric positive SG voltages the QC dip shifts toward positive BG voltages. This effect arises because of the electrostatic influence of the SGs on the GNR channel. A positive SG voltage induces an electrostatic potential in the GNR, leading to an accumulation of electrons and shifting the Fermi level upward, away from the Dirac point. To restore charge neutrality, the Fermi level must be brought back down, which requires a compensating positive BG voltage. The metal BG, being highly conductive and strongly coupled to the GNR through the SiO_2 dielectric, generates a uniform electric field that counteracts the n-doping effect of the SGs. Consequently, the QC dip shifts to higher BG voltages as the system seeks to reestablish charge neutrality.

Additionally a QC peak can be observed at negative BG voltages again contrary to Reiter's results where it occurs at positive BG voltages. This is due to the combined electrostatic effects of SGs and BG on the GNR. Positive V_{SG} shift the Fermi level upward in the SGs, bringing it toward the conduction band but also moving some regions close to the Dirac point. Negative V_{BG} pulls the Fermi level downward in the GNR increasing its capacitive response. At specific combinations of V_{SG} and V_{BG} a region of the SGs reaches the Dirac point, causing the screening effect to collapse.

When the screening effect of the SGs weakens the BG's influence on the GNR increases. This leads to a redistribution of charge in the GNR making the system behave as if it had negative quantum capacitance. This effectively negative quantum capacitance arises because the charge response to an applied voltage is opposite to what is expected, i.e. a decrease of charge accumulation in the GNR with increasing absolute values of V_{BG}. As a result, the total capacitance experienced by the GNR increases, leading to a peak that exceeds the classical limit of $C_{tot} \to C_{Cl}$ for $C_{QM} \to \infty$.

The additional broadening and antisymmetry of the QC dip and peak as the SG voltage increases is due to the non-uniform potential distribution in the GNR. As the SG voltage increases, the SGs cause different regions of the GNR to experience different local electrostatic potentials. Regions closer to the SG experience a stronger potential shift than regions farther away, which causes different parts of the GNR to reach their local Dirac point at different V_{BG} and V_{SG} configurations. Ultimately, this leads to a broader voltage region where the Fermi level is near or at the Dirac point, thus smearing out the otherwise sharp feature of the QC dip and peak.

The asymmetry arises because charge accumulates as a response to V_{SG} in certain areas of the GNR while regions farther away from the SG experience weaker electrostatic influence leading to an asymmetric distribution of charge and asymmetry in the QC dip and peak.

TODO: Insert Contour plots of C_{tot} and α_{rel}

4.4.3 Antisymmetric SG voltages ($V_{SG1} = -V_{SG2}$)

Fig. 26a) depicts the total capacitance per unit area plotted over the BG voltage for the antisymmetric SG voltage configuration, i.e. $V_{SG1} = -V_{SG2}$, for V_{SG1} ranging from 0 V to 7 V. Reiter's results are shown in Fig. 26b) and are again normalized to the respective classical capacitance, hence a quantitative comparison is again not feasible. However, qualitatively the two plots are in good agreement as in both of them a distinct QC dip is observable around 0 V, which becomes broader and thus shallower and more asymmetric with increasing applied SG voltage because of the non-uniform potential in the GNR. Additionally, on both sides of these dips QC peaks can be seen due to reduced SG screening.

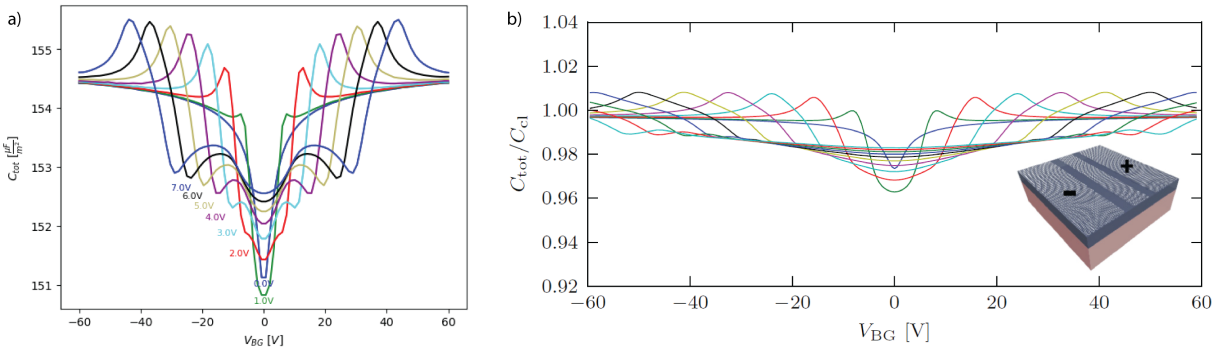


Figure 26: Comparison of (a) the simulation results from this work with (b) the simulation results by Reiter et al. [1], for the antisymmetric voltage configuration of graphene SGs at $w = 80\,\text{nm}$.

The electric field between the SGs creates a strong lateral potential gradient across the GNR. This results in different regions of the GNR experiencing different local electrostatic potentials, meaning that different parts of the GNR reach the Dirac point at slightly different BG voltages. As a consequence, the QC dip broadens.

In an antisymmetric configuration one SG induces an electron-rich region while the other induces a hole-rich region in the GNR. This leads to a partial compensation of charge accumulation, which weakens the reduction in the DOS at the Dirac point resulting in a shallower QC dip.

As the electrostatic potential varies across the GNR the transition from electron to hole conduction occurs unevenly leading to an asymmetric shape in the QC dip. This asymmetry is further enhanced by nonlinear screening effects, i.e. in cedrtain regions the potential is strong while in others it is weaker due to partial compensation by the opposite gate voltage.

The two QC peaks flanking the QC dip arise because of two phenomena. First, a reduced SG screening and thus enhanced BG coupling and second, an effectively negative quantum capacitance. When the SG potential aligns with the Dirac point, the DOS in the SG becomes small and its ability to screen the GNR is significantly reduced. This reduction in screening strengthens the capacitive coupling between BG and GNR.

Thus, at different combinations of V_{SG} and V_{BG} the charge response of the GNR becomes opposite to the expected trend, i.e. instead of charge accumulating with increasing $|V_{BG}|$ the charge accumulation decreases, leading to effectively negative quantum capacitance and a QC peak.

TODO: insert contour plot of C_{tot}

CHAPTER 5

Conclusion

Various graphene- and GNR-FET devices have been simulated using the finite-element software NGSolve. The non-linear Poisson equation was solved self-consistently under the consideration of a potential dependent charge carrier concentration derived via a Thomas-Fermi approach. As a consequence, the accumulated charge in the channel could be calculated and from it the total capacitance. Subsequently, these simulation results were compared with results from the literature [1] for the same geometries using a different approach.

It has been established that our method describes the simple geometry of infinite graphene very well both qualitatively and quantitatively. Furthermore, the results for this geometry are in good agreement with the analytical model derived in [8].

While our results exhibit qualitative agreement with those of Reiter et al. for the more complex geometry, which includes a finite-sized GNR channel, quantitative discrepancies remain. These deviations can be attributed to the assumptions made in deriving the analytical expression for C_{Cl} in [14], which were not fully satisfied in our simulation. Despite this limitation, we adopted this approach because a direct calculation of C_{tot} using our method introduces a small yet non-negligible error, and no alternative analytical expressions for C_{Cl} were available. Reiter et al. used a different approach calculating C_{tot} directly.

For geometries involving SGs, no analytical expression for C_{Cl} was available. Consequently, C_{tot} had to be calculated directly, acknowledging the introduction of a small error. Nevertheless, this approach yielded good quantitative and qualitative agreement with the results of Reiter et al. for metal SGs. Furthermore, the effects of SG screening were clearly observable across all SG voltage configurations.

For graphene side gates, the total capacitance C_{tot} also had to be calculated directly. Our findings reveal the hierarchy $C_{tot}^{NoSG} \gg C_{tot}^{Graphene} > C_{tot}^{Metal}$, reflecting the varying screening capabilities of different side gate materials. Additionally, the QC dip in the grounded metal SG geometry was found to be more pronounced compared to the grounded graphene SG geometry, where the superposition of a QC peak partially diminishes the dip. Furthermore, our results confirm the existence of effectively negative quantum capacitance and the corresponding QC peak observed by Reiter et al. [1]. Overall, the qualitative and quantitative agreement between both simulations is strong across all voltage configurations, with the sole exception of the differing shift behavior of the QC dip in the symmetric SG voltage configuration.

A future work could extend the Poisson solver to support simulations of bilayer graphene and more complex geometries, further improving its applicability and accuracy.

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